Chapter 32 Problem Set

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1. The partial sum s_n is 1 if n is odd and 0 if n is even. So if $\epsilon = 0.5$, no values of m or n satisfy $|s_n - s_m| < \epsilon$. Therefore, the sum diverges.

2. Since $\frac{\log n}{n} \geq \frac{1}{n}$ for $n > e$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, $\sum_{n=1}^{\infty} \frac{\log n}{n}$ diverges too.

4. Since $\frac{\sqrt{n}}{2^n} \leq \frac{\sqrt{n}}{n^{5/2}} = \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{2^n}$ converges too.

5a. Let's say the sum is s. We know that $s_{n+1} = s_n + (-1)^n a_{n+1}$. Since a_n is decreasing, $a_{n+1} \leq a_n$. If n is even, $s_n > s$ and $s_{n+1} < s$. So $s_{n+1} < s < s_n$. If n is odd, then $s_n < s < s_{n+1}$. In both cases, $|s_n - s| \leq a_{n+1}$. Also, since $\lim_{n\to\infty} a_n = 0$ and a_n is positive, for all $\epsilon > 0$ there exists N such that if $n > N$ then $a_n < \epsilon$. This means, given $\epsilon > 0$, there exists N such that if $n > N$ then $|s_n - s| \leq a_{n+1} < \epsilon$.

5b. Let $a_n = \frac{1}{\sqrt{n}}$. We know that $a_i > a_{i+1}$ for all $i \geq 1$ and that $\lim_{n\to\infty} a_n = 0$. By the result of 5a, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ converges.

5c. Since $|s_n - s| \leq a_{n+1}$, if we want 3 digits of accuracy, we want n to be large enough such that $\frac{1}{\sqrt{n+1}} < \frac{1}{1000}$. Isolating n, we get that $n > 999999$, so calculating $s_{1000000}$ should suffice. Doing this, we get 0.6043986435466318.

7a. We see that $\frac{n^n}{5^n+3} \geq \frac{n^n}{8^n} \geq \frac{n}{8}$. Let's say we're given $\epsilon > 0$. If $n > 8\epsilon$, $\frac{n}{8} > \epsilon$, so $\lim_{n\to\infty} \frac{n}{8} \neq 0$ which means $\sum_{n=1}^{\infty} \frac{n}{8}$ diverges, and thus so does $\frac{n^n}{5^n+3}$.

7b. We see that $\frac{n^n}{5^{n^2}} = \frac{n^n}{(5^n)^n} = \left(\frac{n}{5^n}\right)^n < \left(\frac{n}{n^2}\right)^n = \frac{1}{n^n} \leq \frac{1}{n^2}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, $\sum_{n=0}^{\infty} \frac{n^n}{5^{n^2}}$ does too.

7c. We see that $\left(\frac{3n^2+6n+1}{5n^2+1}\right)^n < \left(\frac{3n^2+\frac{3}{5}}{5n^2+1}\right)^n = \left(\frac{3\left(n^2+\frac{1}{5}\right)}{5\left(n^2+\frac{1}{5}\right)}\right)^n = \left(\frac{3}{5}\right)^n$. Since $\sum_{n=0}^{\infty} \left(\frac{3}{5}\right)^n$ converges, $\sum_{n=1}^{\infty} \left(\frac{3n^2+6n+1}{5n^2+1}\right)^n$ does too.

8. Let $a_n = \frac{(-1)^n}{\sqrt{n}}$. Then $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} a_n^2$ doesn't.

Since $\sum_{n=1}^{\infty} a_n$ converges, $\lim_{n\to\infty} a_n = 0$. This implies that there exists N such that for all $n > N$, $a_n < 1$. Multiplying both sides by a_n (we can do this because $a_n > 0$), we have $a_n^2 < a_n$. Therefore, since $\sum_{n=N+1}^{\infty} a_n$ converges, $\sum_{n=N+1}^{\infty} a_n^2$ converges. Also, since $\sum_{n=1}^{N} a_n^2$ is finite, $\sum_{n=1}^{\infty} a_n^2$ also converges.